Checks That Can Fail

Assured AI agents and surrogate models for faster ground-system design and simulation

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1. THE CHALLENGE

AI agents can now carry out whole engineering workflows: they write and revise CAD geometry, set up simulations and write the tests that judge the result, in minutes. Data-driven surrogate models screen a design in microseconds where a physics simulation takes seconds to hours. Both promise faster ground-system development, and both share one weakness: their output is only as good as the checks that accept it. A check, or a surrogate, that passes when it should fail is worse than none. It sends a defect to hardware, where each one costs days, and it erodes trust in the whole pipeline.

I propose to make that acceleration *assured*: to give AI-agent design changes, and the physics and surrogate checks that accept them, measured confidence, while keeping a same-day design, build and test loop. This is the priority because it gates every other use of AI in vehicle engineering, it can be measured, and it transfers across subsystems. It is also an assurance problem of the kind cybersecurity trains for: assume any control can be fooled, by a bug or by the system's own mistake, and test the tests.

2. PRELIMINARY WORK

I have been building **Scorpion**, a tracked ground robot with a 360° pan-tilt camera head, as a platform for embodied AI. Since August 2026 I have run its design in a loop with an AI coding agent (Claude, by Anthropic). I build and test at the bench; the agent turns each finding into a change to parametric, code-based CAD (signed distance fields, CadQuery/OpenCascade), runs automated checks and produces print files, which I print and test, often the same day.

- **Pace.** The pan drive went through 11 design iterations in 16 days; the last four came on consecutive days, each answering the previous day's bench finding.
- **Checks.** Interference at five pan angles, insertion and assembly order, gear backlash, bearing contact, thread engagement and wire clearance; for the whole vehicle, static stability, rest states after a tumble, and which part strikes first over 4,000 fall directions.
- **Finding.** Ten recorded cases of a check that was itself wrong, seven of them passes that let a defect through: parts with zero overlap that could not be assembled, a tumble analysis that scored well while its model was broken, overlapping copies that cancelled in a voxel count. The fix that worked: every check ships with a *negative control*, a deliberately broken variant it must flag, which is mutation testing borrowed from software engineering [1].
- **Escapes.** What still reached hardware was what no check modelled: a material assumption (nylon screws taken for steel), friction, print tolerance and strength. That is where physics models, and data from test, have to come in.

3. PILOT RESULTS

To turn those anecdotes into measurements, the agent and I (below, "we") ran a two-week pilot of each aim on Scorpion in October 2026, under three rules. Each hypothesis and each prediction was committed to version control before its test [7]. Each result has a control, a broken case that must come out differently. Every number here traces to a file in the repository.

- **H1. Mutation-tested checks (Aim 1).** We seeded 22 faults into the pan drive's model, 14 of them re-creations of real failures from its history (Fig. 1A). The project's first check, part-against-part interference, caught 12 (7 of the 14 real ones). The suite grown since, one check for each escape, caught 21 (all 14). The one survivor, a part sitting 0.6 mm out of its grooves, became a new check: 22 of 22. None of the ten checks is vacuous; each catches at least one fault.

- **H2. An uncertainty-gated surrogate (Aim 2).** A MuJoCo simulation [3] tumbled 400 body shapes 200 times each, 80,000 tumbles in 74 s on 20 cores, to find which shapes come to rest upright. A Gaussian process [5] trained on 120 of them predicted the other 280 with an RMS error of 3.9 percentage points, and its 90% intervals held 86.8% of the time (Fig. 1B). Asked whether each design lands upright at least 60% of the time, it answered 266 of 280 by itself, 2 of them wrongly, and sent 14 to simulation: 95% of the simulations saved. Two checks exposed its limits, and both are findings. A registered control failed: sentinel runs could not see a hidden friction change, which moved the answer by at most 4 points. And a later check found that intervals that held on average held for only 1 of the 6 best designs, all about 5 points too low: exactly where a design gets chosen.
- **H3. Simulation against reality.** We printed scale models of three candidate bodies, each with a coin pocket so its centre of mass can be moved as a physical control, and I dropped them by hand 361 times, from 6 and 30 cm. Each prediction is scored by its Brier score [6]: 0 is perfect, and lower is better (Fig. 1C). At 6 cm the committed predictions beat a naive guess, 0.58 against 1.29, and shifted the right way when the coins moved. That pass was partly luck. Breaking the misses down by starting position exposed a bug in how simulated drops began, and fixing it alone made the score worse (0.86). Watching drops in the simulator's viewer, I saw its default floor let bodies sink 7–17 mm on impact, which a passing audit had measured only at rest. A floor bounce fitted at 6 cm and committed before the 30 cm drops then won that blind test: 0.63 against 1.16 for the default floor (difference 0.52, 95% CI 0.40–0.64). After the fact, a hard floor with its bounce measured on 240 fps video comes close without fitting anything: 0.46 and 0.67, where the drops' own randomness sets a best possible score near 0.44 and 0.60. The model still over-predicts how much moving the weight matters at 30 cm. A third blind test, at 15 cm, [result].
- **H4. Cycle time.** Reconstructed from version control and logs (retrospective), the median from a finding to a fix ready to print was 1.6 h (interquartile range 0.6–2.5 h, $n = 20$; 1.9 h for bench findings); 10 of 26 findings were caught before printing, and 5 had slipped past a check that passed. During the pilot, 15 more findings were logged as they happened, with commit times.

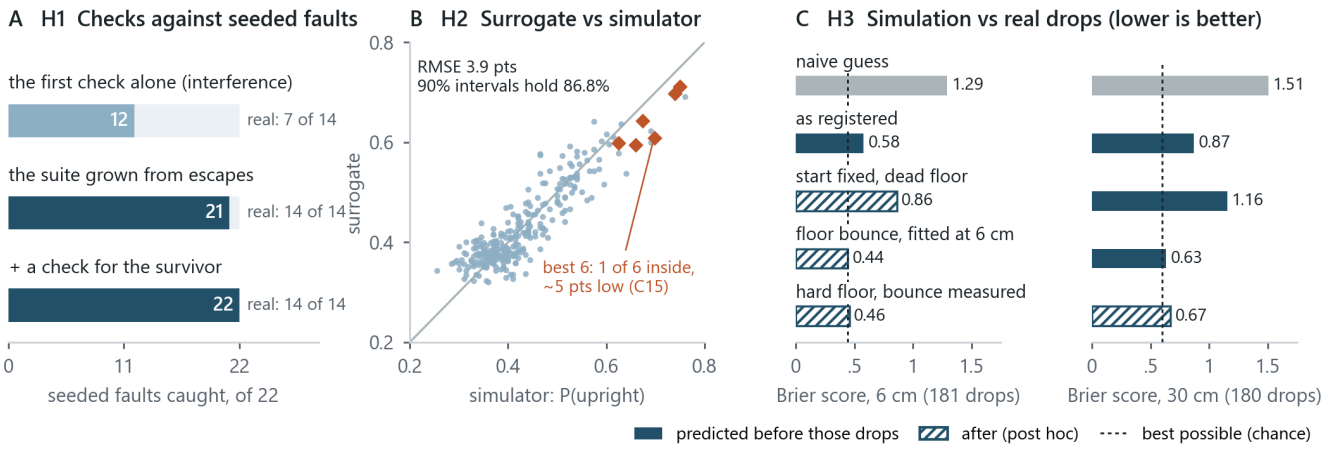


Figure 1. Pilot results. (A) Seeded faults caught by the first check alone, by the suite grown from escapes, and after the survivor became a check; "real" counts the 14 re-created historical failures. (B) Surrogate against simulator on 280 held-out designs; orange, the 6 best designs, each predicted without its own data, against 1,000-drop reruns. (C) Brier scores against the real drops: solid bars committed before those drops, hatched after (the 6 cm floor bounce was fitted to them); dashed, the best score the drops' own randomness allows.

What the pilot shows. The project has now recorded 15 cases of a check that was itself wrong, 12 of them passes that should have failed, and 5 of them during the pilot: in a CAD check suite, the drop predictions, the simulator's floor, a numerical time step and the surrogate's intervals. None was caught by the check itself. Each was exposed by something outside it: a person looking, a physical test, a cross-check or a control, and each changed how that kind of check is done. That is the case for this plan: checks that can fail, and are shown to.

4. APPROACH

Aim 1: Mutation-tested agent workflows. Grow the pilot's harness into a library of fault operators (move a part, close a clearance, swap a material, sever a joint, duplicate a body, perturb a load) that scores each check by the faults it catches. The agent must deliver every design change with its checks and their mutants, and a change is accepted only when its checks reach a set mutation score. Designs, results and test logs are versioned and signed: a chain of custody for AI-generated engineering artifacts.

Aim 2: Physics, surrogates and uncertainty. Couple the checks to physics-based simulation: Project Chrono for vehicle-terrain mobility and rollover [2], MuJoCo for mechanisms [3], and [Org] and SPARC tools where they fit. Train surrogates on simulation sweeps run on high-performance computing, so the agent can screen a change in seconds; send any case a surrogate is unsure of to full simulation or physical test [4]; and check its intervals where decisions are made, not only on average. A surrogate is a check too, so it faces the same mutants. Bench data calibrates the physics, and each escape becomes both a regression case and new training data.

Aim 3: Demonstrate and measure. With the Chief Scientist, choose a ground-system problem from [Org] and SPARC priorities, for example a sensor mast or turret mount, a drive module, or rollover screening. Run a controlled comparison on matched tasks: the conventional workflow, AI agents alone, and AI agents with mutation-tested, uncertainty-gated checks. Measures: time from a test finding to a verified change; defects that reach hardware; mutation score; whether the surrogate's stated uncertainty holds; and compute and engineer hours.

5. SCHEDULE (12 MONTHS)

Months 1-2	Evaluate SPARC foundation models, frameworks and simulation codes against the three aims; select the demonstration problem; package the pilot's toolkit and fault-operator library.
Months 3-6	Physics coupling and surrogate training on HPC; uncertainty gating; bench calibration; provenance logging.
Months 7-10	Controlled study on the selected subsystem.
Months 11-12	Analysis, transition package, technical report and publication.

6. DELIVERABLES, TRANSITION AND PARTNERSHIP

An open check-and-mutation library, in Python on open formats (STEP, STL, MJCF), that drops into existing CAD and simulation pipelines; trained surrogates with their validation records; a workflow guide and a half-day course for [Org] engineers; a dataset of design iterations, escapes and cycle times; a technical report and a paper. I will pair [Org] staff with UM and LANL experts in scientific machine learning and verification, and draft at least one joint follow-on proposal. Transition succeeds when a second team runs the harness on its own vehicle without me.

7. RISKS

- **Simulation and surrogate fidelity.** The pilot found both wrong in ways only outside checks exposed. Physical test stays the arbiter, and uncertainty gating limits where a surrogate may decide alone.
- **Agent errors.** No change is accepted without passing its mutants and a human review gate.
- **Scale.** Shown on a small robot so far; Aim 3 tests it on an [Org] subsystem, with its own tools.

REFERENCES

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